

1.a) What are the advantages of digital signal processing over analog signal processing? Briefly explain the basic elements of digital signal processing with the help of block diagram.

sol: Digital signal processing is the process of processing discrete signals that are sampled and represented as numbers.

The advantages of digital signal processing over analog signal processing are as follows:

(i) Flexibility:

→ The digital signal processing systems are flexible because it can be changed by just updating the software.

(ii) Accuracy:

→ The accuracy of digital signal processing systems is more higher than analog systems.

(iii) Storage:

→ The digital signals can be easily stored on the storage media such as magnetic tapes, disks etc.

(iv) Mathematical Processing:

→ Mathematical operations can be accurately performed on digital signals as compared to analog signals.

(v) Cost:

→ When there is large complexity in the application, then digital signal processing systems are cheaper compared to analog systems.

(vi) Adaptability:

→ The digital signal processing systems are easily upgradable ~~since~~ since they are software controlled.

(vii) Size and Reliability:

→ The digital signal processing systems are small in size and more reliable compared to analog systems.

The basic elements of digital signal processing is shown below:

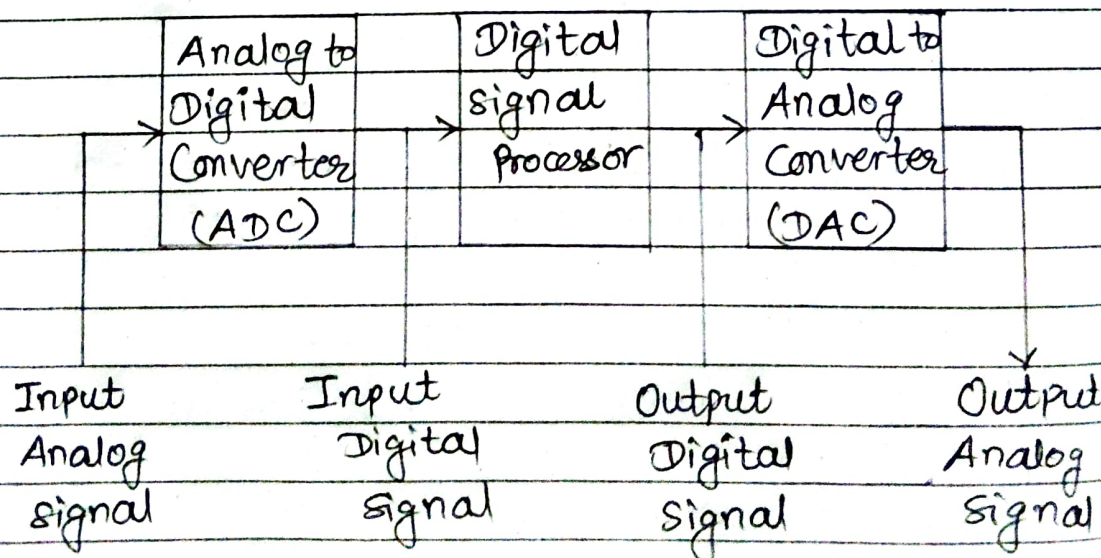


Fig. Basic Elements of DSP

### (i) Analog to Digital Converter (ADC):

→ Initially, the input analog signal passes through the analog to digital converter (ADC) to produce the input analog signal which is digital appropriate for the digital signal processor.

### (ii) Digital Signal Processor:

→ The digital signal processor is the large programmable digital computer which converts the input digital signal to the output digital signal.

### (iii) Digital to Analog Converter (DAC):

→ Some of the processed signals are required to back in their original form i.e. analog form, so the output of the digital signal processor is passed through the digital to analog converter (DAC) which converts the output digital signal to its equivalent analog form.

1.b) State and prove the necessary and suff

1.b) Obtain a linear convolution of the following two discrete-time signals.

$$x(n) = \sum_{k=0}^3 \delta(n-k) \text{ and}$$

$$h(n) = 2^n [u(n) - u(n-3)]$$

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Here,

$$x(n) = \sum_{k=0}^3 \delta(n-k) \text{ and}$$

$$h(n) = 2^n [u(n) - u(n-3)]$$

when  $n=0$ :

$$\begin{aligned} x(0) &= \delta(0-0) + \delta(0-1) + \delta(0-2) + \delta(0-3) \\ &= \delta(0) + \delta(-1) + \delta(-2) + \delta(-3) \\ &= 1 + 0 + 0 + 0 \end{aligned}$$

$$\therefore x(0) = 1$$

$$\begin{aligned} h(0) &= 2^0 [u(0) - u(0-3)] = 1 [1 - 0] \\ &= 1 \end{aligned}$$

when  $n=1$ :

$$\begin{aligned} x(1) &= \delta(1-0) + \delta(1-1) + \delta(1-2) + \delta(1-3) \\ &= \delta(1) + \delta(0) + \delta(-1) + \delta(-2) \\ &= 0 + 1 + 0 + 0 \\ &= 1 \end{aligned}$$

$$\begin{aligned} h(1) &= 2^1 [u(1) - u(1-3)] = 2 [1 - 0] \\ &= 2 \end{aligned}$$

when  $n=2$ :

$$\begin{aligned} x(2) &= \delta(2-0) + \delta(2-1) + \delta(2-2) + \delta(2-3) \\ &= \delta(2) + \delta(1) + \delta(0) + \delta(-1) \\ &= 0 + 0 + 1 + 0 \\ &= 1 \end{aligned}$$

$$h(2) = 2^2 [u(2) - u(2-3)] = 4(1-0) = 4$$

when  $n=3$ :

$$\begin{aligned} x(3) &= \delta(3-0) + \delta(3-1) + \delta(3-2) + \delta(3-3) \\ &= \delta(3) + \delta(2) + \delta(1) + \delta(0) \\ &= 0 + 0 + 0 + 1 \\ &= 1 \end{aligned}$$

$$h(3) = 2^3 [u(3) - u(3-3)] = 4(1-1) = 0$$

Hence,

$$\begin{aligned} x(n) &= \overset{\downarrow}{\{1, 1, 1, 1\}} \\ h(n) &= \underset{\uparrow}{\{1, 2, 4\}} \end{aligned}$$

We have,

$$\text{Length of } x(n) = L = 4$$

$$\text{Length of } h(n) = M = 3$$

$$\therefore \text{Length of } y(n) = L + M - 1 = 4 + 3 - 1 = 6$$

$$\therefore \text{Lower limit of } y(n) [K] = 0 + 0 = 0$$

$$\therefore \text{Upper limit of } y(n) [K] = 3 + 2 = 5$$

The linear convolution of  $y(n)$  is given by

$$y(n) = \sum_{k=0}^5 x(k) h(n-k)$$

$$y(n) = x(0)h(n-0) + x(1)h(n-1) + x(2)h(n-2) + x(3)h(n-3) + x(4)h(n-4) + x(5)h(n-5)$$

For  $n=0$ :

$$\begin{aligned} y(0) &= x(0)h(0) + x(1)h(-1) + x(2)h(-2) + x(3)h(-3) \\ &\quad + x(4)h(-4) + x(5)h(-5) \\ &= 1 \cdot 1 + 0 + 0 + 0 + 0 + 0 \\ &= 1 \end{aligned}$$

For  $n=1$ :

$$\begin{aligned} y(1) &= x(0)h(1) + x(1)h(0) + x(2)h(-1) + x(3)h(-2) \\ &\quad + x(4)h(-3) + x(5)h(-4) \\ &= 1 \cdot 2 + 1 \cdot 1 + 0 + 0 + 0 + 0 \\ &= 3 \end{aligned}$$

For  $n=2$ :

$$\begin{aligned} y(2) &= x(0)h(2) + x(1)h(1) + x(2)h(0) + x(3)h(-1) \\ &\quad + x(4)h(-2) + x(5)h(-3) \\ &= 1 \cdot 4 + 1 \cdot 2 + 1 \cdot 1 + 0 + 0 + 0 \\ &= 7 \end{aligned}$$

For  $n=3$ :

$$\begin{aligned} y(3) &= x(0)h(3) + x(1)h(2) + x(2)h(1) + x(3)h(0) \\ &\quad + x(4)h(-1) + x(5)h(-2) \\ &= 1 \cdot 0 + 1 \cdot 4 + 1 \cdot 2 + 1 \cdot 1 + 0 + 0 \\ &= 7 \end{aligned}$$

For  $n=4$ :

$$\begin{aligned} y(4) &= x(0)h(4) + x(1)h(3) + x(2)h(2) + x(3)h(1) \\ &\quad + x(4)h(0) + x(5)h(-1) \\ &= 1 \cdot 0 + 1 \cdot 0 + 1 \cdot 4 + 1 \cdot 2 + 0 \cdot 1 + 0 \\ &= 6 \end{aligned}$$

For  $n=5$ :

$$\begin{aligned} y(5) &= x(0)h(5) + x(1)h(4) + x(2)h(3) + x(3)h(2) \\ &\quad + x(4)h(1) + x(5)h(0) \\ &= 1 \cdot 0 + 1 \cdot 0 + 1 \cdot 0 + 1 \cdot 4 + 0 \cdot 2 + 0 \cdot 1 \\ &= 4 \end{aligned}$$

Hence, the required convolution is:

$\{1, 3, 7, 7, 6, 4\}$  Ans

2.a) Define the term causality, stability, time-invariance and linearity of discrete time system. Also state their significance in case of LTI system.

Sol:

(i) Causality:

→ A discrete-time system is causal if the present output depends only on the present and past inputs, but not on the future inputs.

Mathematically,

$$y(n) = T\{x(n)\}$$

$y(n_0)$  depends only on  $x(n)$  for  $n \leq n_0$ .

\* Significance in LTI system;

→ Causality ensures that the system can be physically realized in real-time, because future inputs are not available.

(ii) Stability:

→ A discrete-time system is BIBO stable if a bounded input always produces a bounded output.

Mathematically,

$$|x(n)| \leq M_x < \infty \Rightarrow |y(n)| \leq M_y < \infty$$

\* Significance in LTI system;

→ Stability guarantees that the bounded outputs for bounded inputs.

## (iii) Time-Invariance:

→ A system is said to be time-invariant if a time shift in the input results in an identical time shift in the output.  
Mathematically,

If  $y(n) = T\{x(n)\}$ , Then,

$$T\{x(n-n_0)\} = y(n-n_0)$$

\* Significance in LTI systems:

→ Time-invariance ensures that the system's behavior does not change over time, making it predictable.

## (iv) Linearity:

→ A discrete-time system is linear if it satisfies:

• Additivity:

$$T\{x_1(n) + x_2(n)\} = Tx_1(n) + Tx_2(n)$$

• Homogeneity:

$$T[ax(n)] = a[Tx(n)] \text{ for any } a.$$

\* Significance in LTI systems:

→ Linearity allows the use of superposition and convolution, simplifying analysis and design.

2.6) Define z-transform. Find the inverse z transform of  $X(z) = \log(1+az^{-1})$ .

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The z-transform is the discrete-time counterpart of the Laplace transform.

The z-transform of a discrete time signal  $x(n)$  is defined as:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

The inverse z-transform of  $X(z)$  is defined as:

$$x(n) = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$

We have,

$$X(z) = \log(1+az^{-1}) \text{ with ROC: } |z| > |a|.$$

On differentiating, we get:-

$$\frac{d}{dz} (X(z)) = \frac{d}{dz} [\log(1+az^{-1})] ; |z| > |a|$$

$$\Rightarrow \frac{d}{dz} (X(z)) = \frac{1}{(1+az^{-1})} \cdot (-az^{-2}) ; |z| > |a|$$

$$\Rightarrow \frac{d}{dz} [X(z)] = \frac{-az^{-2}}{1+az^{-1}} ; |z| > |a|$$

$$\Rightarrow \frac{d}{dz} [x(z)] = \frac{-az^{-1}}{z(1+az^{-1})}; |z| > |a|$$

$$\Rightarrow -z \frac{d}{dz} [x(z)] = \frac{+az^{-1}}{1+az^{-1}}; |z| > |a|$$

Applying differentiation property on LHS and taking inverse z-transform on both sides,

$$n x(n) = z^{-1} \left[ \frac{az^{-1}}{1+az^{-1}} \right]$$

( $\because$  By time shifting property on RHS)

$$\Rightarrow n x(n) = a(-a)^{n-1} u(n-1)$$

$$\Rightarrow x(n) = \frac{1}{n} a(-a)^{n-1} u(n-1)$$

$$\Rightarrow x(n) = \frac{a(-a)^{n-1}}{n(-a)} u(n-1)$$

$$\Rightarrow x(n) = \frac{(-1)^1 \cdot (-1)^n a^n}{n} u(n-1)$$

$$\therefore x(n) = \frac{(-1)^{n+1} a^n}{n} u(n-1)$$

This is the required inverse z-transform of the given. Ans

Rough:

We know,

By time shifting property

~~we know~~

$$a^n u(n) \longleftrightarrow \frac{1}{1-az^{-1}}; |z| > |a|$$

$$(-a)^n u(n) \longleftrightarrow \frac{1}{1+az^{-1}}; |z| > |a|$$

$$a \cdot (-a)^n u(n) \longleftrightarrow \frac{a}{1+az^{-1}}; |z| > |a|$$

$$\therefore a(-a)^{n-1} u(n-1) \longleftrightarrow \frac{az^{-1}}{1+az^{-1}}; |z| > |a|$$

3.a) Find the 8-point DIFFFT of  $x(n) = \sin \frac{\pi n}{3}$ .

Here,

$$x(n) = \sin \frac{\pi n}{3}$$

$$\text{When } n=0; x(0) = \sin \frac{\pi \times 0}{3} = 0$$

$$\text{when } n=1; x(1) = \sin \frac{\pi \times 1}{3} = 0.87$$

$$\text{when } n=2; x(2) = \sin \frac{\pi \times 2}{3} = 0.87$$

$$\text{when } n=3; x(3) = \sin \frac{\pi \times 3}{3} = 0$$

$$\text{when } n=4; x(4) = \sin \frac{\pi \times 4}{3} = -0.87$$

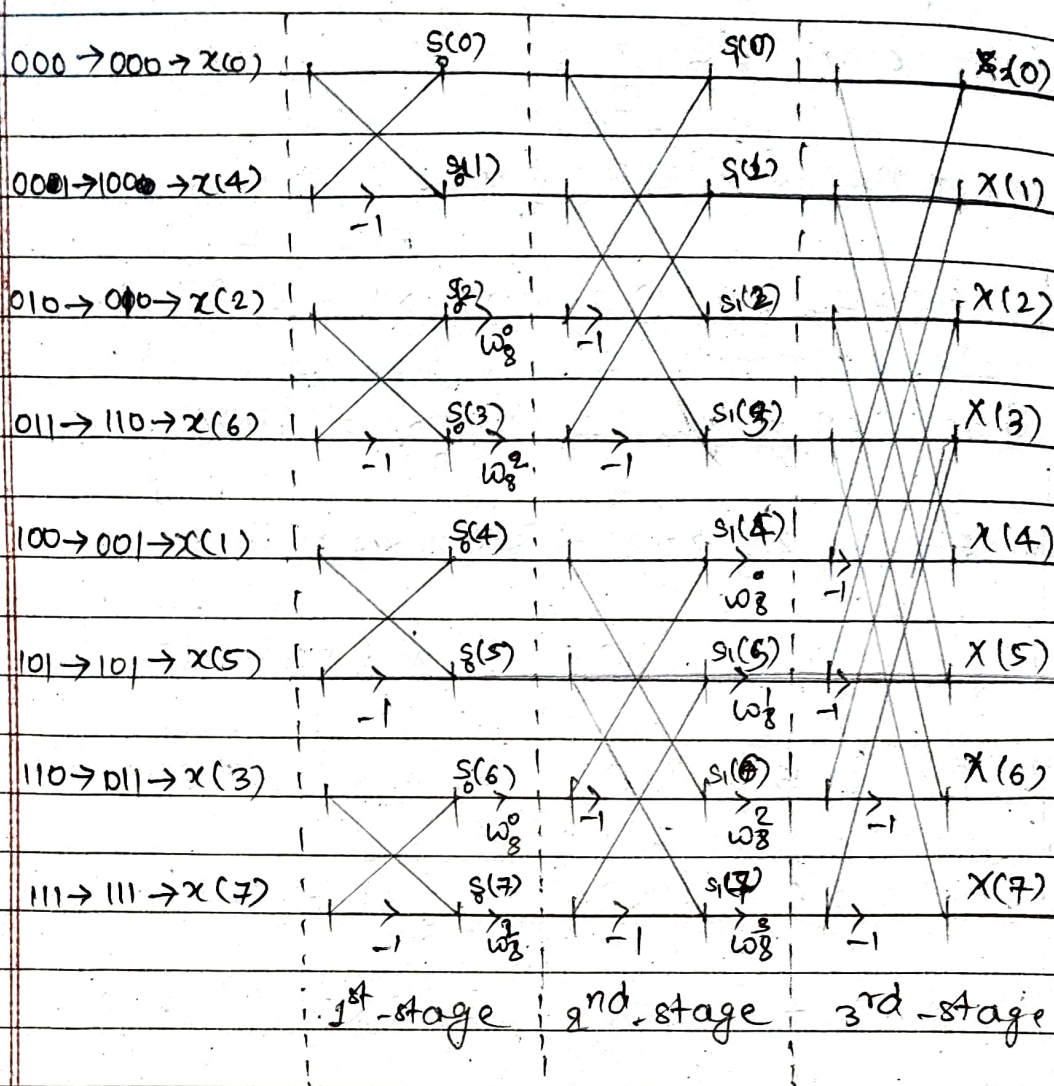
$$\text{when } n=5; x(5) = \sin \frac{\pi \times 5}{3} = -0.87$$

$$\text{when } n=6; x(6) = \sin \frac{\pi \times 6}{3} = 0$$

$$\text{when } n=7; x(7) = \sin \frac{\pi \times 7}{3} = 0.87$$

$$\text{Hence, } x(n) = \{0, 0.87, 0.87, 0, -0.87, -0.87, 0, 0.87\}$$

The flow graph is shown below:



i We have,

$$w_8^0 = e^{-j2\pi/8} = e^0 = 1$$

$$w_8^1 = e^{-j\pi/8} = e^{-j\pi/8} = \cos\left(\frac{\pi}{8}\right) - j \sin\left(\frac{\pi}{8}\right)$$

$$= 0.707 - j0.707$$

$$w_8^2 = e^{-j2\pi/8} = e^{-j4\pi/8} = -j$$

$$w_8^3 = e^{-j3\pi/8} = -0.707 - j0.707$$

(i) Output of 1<sup>st</sup> stage:

$$S_0(0) = x(0) + x(4) = 0 + -0.87 = -0.87$$

$$S_0(1) = x(0) + x(4) x^{-1} = 0 + -0.87 x^{-1} = 0.87$$

$$S_0(2) = x(2) + x(6) = 0.87 + 0 = 0.87$$

$$S_0(3) = x(2) + x(6) x^{-1} = 0.87 + 0 x^{-1} = 0.87$$

$$S_0(4) = x(1) + x(5) = 0.87 + -0.87 = 0$$

$$S_0(5) = x(1) + x(5) x^{-1} = 0.87 + -0.87 x^{-1} = 1.74$$

$$S_0(6) = x(3) + x(7) = 0 + 0.87 = 0.87$$

$$S_0(7) = x(3) + x(7) x^{-1} = 0 + 0.87 x^{-1} = -0.87$$

(ii) Output for 2<sup>nd</sup> stage:

$$S_1(0) = S_0(0) + S_0(2) \times \omega_8^0 = -0.87 + 0.87 \times 1 = 0$$

$$\begin{aligned} S_1(1) &= S_0(1) + S_0(3) \times \omega_8^2 = 0.87 + 0.87 \times j \\ &= 0.87 - j0.87 \end{aligned}$$

$$\begin{aligned} S_1(2) &= S_0(0) + S_0(2) \times \omega_8^0 \times -1 = -0.87 + 0.87 \times 1 \times -1 \\ &= -1.74 \end{aligned}$$

$$\begin{aligned} S_1(3) &= S_0(1) + S_0(3) \times \omega_8^2 \times -1 = 0.87 + 0.87 \times j \times -1 \\ &= 0.87 + j0.87 \end{aligned}$$

$$S_1(4) = S_0(4) + S_0(6) \times \omega_8^0 = 0 + 0.87 \times 1 = 0.87$$

$$\begin{aligned} S_1(5) &= S_0(5) + S_0(7) \times \omega_8^2 = 1.74 + -0.87 \times j \\ &= 1.74 + j0.87 \end{aligned}$$

$$\begin{aligned} S_1(6) &= S_0(4) + S_0(6) \times \omega_8^0 \times -1 = 0 + 0.87 \times 1 \times -1 \\ &= -0.87 \end{aligned}$$

$$\begin{aligned} S_1(7) &= S_0(5) + S_0(7) \times \omega_8^2 \times -1 = 1.74 + -0.87 \times j \times -1 \\ &= 1.74 - j0.87 \end{aligned}$$

(iii) Output For Anal stage:

$$X(0) = S_1(0) + S_1(4) \times \omega_8^0 = 0 + 0.87 \times 1 = 0.87$$

$$X(1) = S_1(1) + S_1(5) \times \omega_8^1 = (0.87 - j0.87) + (1.74 + j0.87) \times (0.707 - j0.707) = 2.71 - j1.48$$

$$X(2) = S_1(2) + S_1(6) \times \omega_8^2 = (-1.74) + (-0.87) \times -j = -1.74 + j0.87$$

$$X(3) = S_1(3) + S_1(7) \times \omega_8^3 = (0.87 + j0.87) + (1.74 - j0.87) \times (-0.707 - j0.707) = -0.97 + j0.25$$

$$X(4) = S_1(0) + S_1(4) \times \omega_8^0 \times -1 = 0 + 0.87 \times 1 \times -1 = -0.87$$

$$X(5) = S_1(1) + S_1(5) \times \omega_8^1 \times -1 = (0.87 - j0.87) + (1.74 + j0.87) \times (0.707 - j0.707) \times -1 = -0.97 - j0.25$$

$$X(6) = S_1(2) + S_1(6) \times \omega_8^2 \times -1 = (-1.74) + (-0.87) \times -j \times -1 = -1.74 - j0.87$$

$$X(7) = S_1(3) + S_1(7) \times \omega_8^3 \times -1 = (0.87 + j0.87) + (1.74 - j0.87) \times (-0.707 - j0.707) \times -1 = 2.71 + j1.48$$

Hence the required 8-point DIFFFT are as :-

$$X(n) = \{0.87, 2.71 - j1.48, -1.74 + j0.87, -0.97 + j0.25, -0.87, -0.97 - j0.25, -1.74 - j0.87, 2.71 + j1.48\}$$

Ans

3.6) Show that the multiplication of two DFT sequences results in circular convolution.

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Let us consider two discrete time sequence  $x_1(n)$  and  $x_2(n)$ . The DFT of  $x_1(n)$  and  $x_2(n)$  can be expressed as,

$$X_1(K) = \sum_{n=0}^{N-1} x_1(n) e^{-j2\pi Kn} \quad \text{--- (1)}$$

where,  $K = 0, 1, \dots, N-1$

$$X_2(K) = \sum_{l=0}^{N-1} x_2(l) e^{-j2\pi Kl} \quad \text{--- (2)}$$

where,  $K = 0, 1, \dots, N-1$

From (1) & (2), we get;

$$Y(K) = X_1(K) \cdot X_2(K) \quad \text{--- (3)}$$

let IDFT of  $Y(K)$  be  $y(m)$ . Then using definition of IDFT, we have

$$y(m) = \frac{1}{N} \sum_{K=0}^{N-1} Y(K) e^{j2\pi Km} \quad \text{--- (4)}$$

From (3) & (4), we get;

$$y(m) = \frac{1}{N} \sum_{K=0}^{N-1} X_1(K) X_2(K) e^{j2\pi Km} \quad \text{--- (5)}$$

From (1), (2) & (5), we get:-

$$y(m) = \frac{1}{N} \sum_{k=0}^{N-1} \left[ \sum_{n=0}^{N-1} x_1(n) e^{-j \frac{2\pi k n}{N}} \right]$$

$$\left[ \sum_{l=0}^{N-1} x_2(l) e^{-j \frac{2\pi k l}{N}} \right] \cdot e^{j \frac{2\pi k m}{N}}$$

$$\Rightarrow y(m) = \frac{1}{N} \sum_{k=0}^{N-1} x_1(n) \sum_{l=0}^{N-1} x_2(l)$$

$$\left[ \sum_{k=0}^{N-1} e^{-j \frac{2\pi k l}{N}} \cdot e^{-j \frac{2\pi k n}{N}} \cdot e^{j \frac{2\pi k m}{N}} \right]$$

$$\therefore y(m) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \sum_{l=0}^{N-1} x_2(l) \left[ \sum_{k=0}^{N-1} e^{j \frac{2\pi k (m-n-l)}{N}} \right]$$

———— (6)

Let us consider the last term of equation. It may be written as under

$$e^{j \frac{2\pi k (m-n-l)}{N}} = \left[ e^{j \frac{2\pi (m-n-l)}{N}} \right]^k \quad \text{--- (7)}$$

Let us use the following standard summation expression;

$$\sum_{k=0}^{N-1} a^k = \begin{cases} N & ; a = 1 \\ \frac{1-a^N}{1-a} & ; a \neq 1 \end{cases}$$

where  $a = e^{j \frac{2\pi (m-n-l)}{N}}$

(i) case (i): When  $\alpha = 1$ :

If  $(m-n-l)$  is multiple of  $N$  which means  
 $(m-n-l) = N, 2N, 3N, \dots$

$$a = e^{+j2\pi} = e^{+j2\pi(2)} = e^{+j2\pi(3)} \dots = 1$$

(ii) case (ii): When  $\alpha \neq 1$ :

If  $\alpha \neq 1$ , this means that if  $m-n-l$  is not multiple of  $N$  then according to equation, we have,

$$\sum_{k=0}^{N-1} \alpha^k = \frac{1-\alpha^N}{1-\alpha} \quad \text{--- (8)}$$

From (7) & (8), we get:-

$$\sum_{k=0}^{N-1} [e^{+j2\pi(m-n-l)}]^k = \frac{1 - e^{+j2\pi(m-n-l)}}{1 - e^{+j2\pi(m-n-l)/N}}$$

where  $m, n$  &  $l$  are integers.

From (6),

~~$$y(m) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \sum_{l=0}^{N-1} x_2(l) \cdot N$$~~

$$\therefore y(m) = \sum_{n=0}^{N-1} x_1(n) \cdot \sum_{l=0}^{N-1} x_2(l) \quad \text{--- (9)}$$

If  $p$  is an integer, which can be positive or negative. So,

$$m - n - l = -pN$$

$$\therefore l = m - n + pN$$

From (9),

$$y(m) = \sum_{n=0}^{N-1} x_1(n) \cdot x_2(m - n + pN)$$

$$\Rightarrow y(m) = \sum_{n=0}^{N-1} x_1(n) \cdot x_2((m - n))_N$$

$$\therefore y(m) = \sum_{n=0}^{N-1} x_1(n) \cdot x_2((m - n))_N ; m = 0, 1, \dots, N-1$$

This shows that the multiplication of two DFT sequences results in circular convolution.

4.a) Find the direct form-I and direct form-II realizations of a discrete-time system represented by the transfer function

$$H(z) = \frac{3z^3 - 5z^2 + 9z - 3}{\left[z - \left(\frac{1}{2}\right)\right]\left[z^2 - z + \left(\frac{1}{3}\right)\right]}$$

sol Here,

$$H(z) = \frac{3z^3 - 5z^2 + 9z - 3}{\left[z - \left(\frac{1}{2}\right)\right]\left[z^2 - z + \left(\frac{1}{3}\right)\right]}$$

$$= \frac{3z^3 - 5z^2 + 9z - 3}{\left(\frac{z^2 - 1}{z}\right)\left(z^2 - z + \frac{1}{3}\right)}$$

$$= \frac{z(3z^3 - 5z^2 + 9z - 3)}{(z^2 - 1)\left(z^2 - z + \frac{1}{3}\right)}$$

$$= \frac{3z^4 - 5z^3 + 9z^2 - 3z}{z^4 - z^3 - \frac{2}{3}z^2 + z - \frac{1}{3}}$$

$$= \frac{z^4(3 - 5z^{-1} + 9z^{-2} - 3z^{-3})}{z^4\left(1 - z^{-1} - \frac{2}{3}z^{-2} + z^{-3} - \frac{1}{3}z^{-4}\right)}$$

$$\therefore H(z) = \frac{3 - 5z^{-1} + 9z^{-2} - 3z^{-3}}{1 - z^{-1} - \frac{2}{3}z^{-2} + z^{-3} - \frac{1}{3}z^{-4}} \quad (1)$$

Comparing w with

$$H(z) = \frac{B(z)}{A(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + b_3 z^{-3}}{a_0 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3} + a_4 z^{-4}}$$

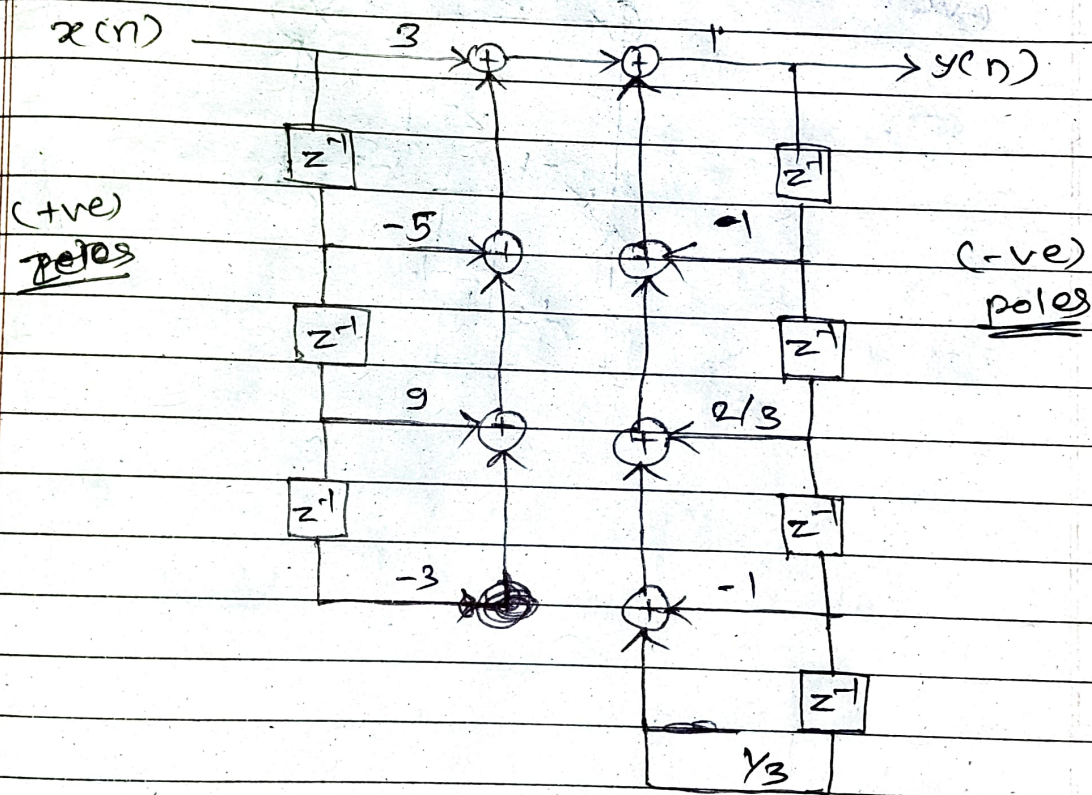
we get;

$$b_0 = 3; b_1 = 5; b_2 = 9; b_3 = -3$$

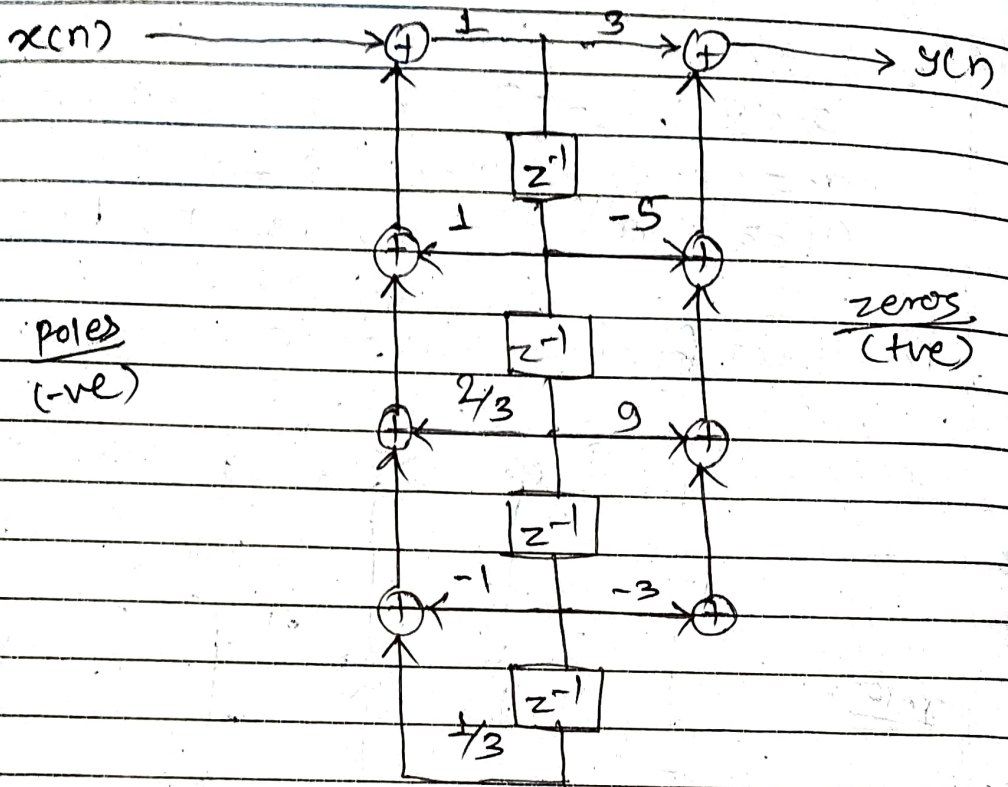
$$a_0 = 1 \text{ (i.e. } a_0 \text{ must be 1)}$$

$$a_1 = -1; a_2 = -2/3; a_3 = 1; a_4 = -1/3$$

The direct form I structure is shown below:



The direct form II structure is shown below:



4.6) How are FIR digital filters designed using different approaches? How would you use a rectangular window to design a FIR filter? Explain.

Ans FIR digital filters are designed using different approaches by following ways:-

- (i) Window method (Linear Phase)
- (ii) Frequency Sampling Method
- (iii) Gibb's Phenomena

We would use a rectangular window to design a FIR filter by following:-

Let us consider that the digital filter which is to be designed have the frequency response  $H_d(\omega)$ . This is also called ~~dire~~ desired frequency response.

Let  $h_d(n)$  be the corresponding unit sample response. We know,

$$H_d(\omega) = \sum_{n=0}^{\infty} h_d(n) e^{-j\omega n} \quad (1)$$

The Inverse Fourier Transform of  $H_d(\omega)$  is given by

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega \quad (2)$$

Usually,  $h_d(n)$  in eqn (2) is infinite in duration. Since, we are designing a finite impulse response filter, the length of  $h_d(n)$  must be made finite.

If we want the unit sample of response of length  $M$  then  $h_d(n)$  is truncated to length  $M$ . This is equivalent to multiply  $h_d(n)$  by a window function  $w(n)$ .

$$\text{i.e. } h(n) = h_d(n) \cdot w(n) \quad \text{--- (3)}$$

For rectangular window,

$$w_R(n) = \begin{cases} 1 & ; n = 0, 1, \dots, M-1 \\ 0 & ; \text{otherwise} \end{cases}$$

From (3),

$$\therefore h(n) = h_d(n) \cdot w_R(n) \quad \text{--- (4)}$$

~~Since~~, from (4), we got,

$$h(n) = \begin{cases} h_d(n) & ; n = 0, 1, \dots, M-1 \\ 0 & ; \text{otherwise} \end{cases}$$

Hence  $h_d(n)$  is truncated to length  $M$  to give  $h(n)$ .

The magnitude response of rectangular window is given by

$$|w_R(\omega)| = \frac{\left| \sin\left(\frac{\omega M}{2}\right) \right|}{\left| \sin\left(\frac{\omega}{2}\right) \right|}$$

5.a) Design a linear FIR filter using Kaiser window to meet the following specifications:

$$0.99 \leq |H(e^{j\omega})| \leq 1.01; \text{ for } 0 \leq |\omega| \leq 0.19\pi$$

$$\leq |H(e^{j\omega})| \leq 0.01; \text{ for } 0.21\pi \leq |\omega| \leq \pi$$

891: Here,  $1+\delta_1$  and  $1-\delta_1$  are ripple factors.

from above specifications,

$$1+\delta_1 = 1.01 \quad \therefore \delta_1 = 0.01$$

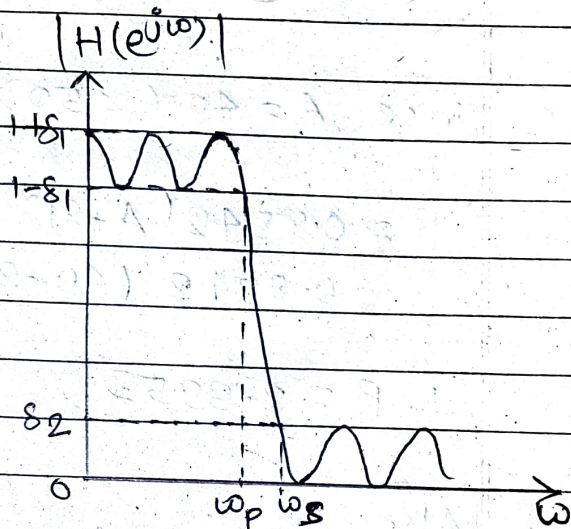
$$1-\delta_1 = 0.99 \quad \therefore \delta_1 = 0.01$$

We got;

$$\omega_p = 0.19\pi$$

$$\omega_s = 0.21\pi$$

$$\delta_2 = \delta_1 = 0.01$$



$$\therefore \text{Transition width, } \Delta\omega = \omega_s - \omega_p = 0.21\pi - 0.19\pi$$

$$= 0.02\pi$$

$$\therefore \text{Cut-off frequency, } \omega_c = \frac{\omega_p + \omega_s}{2} = \frac{0.19\pi + 0.21\pi}{2}$$

$$= 0.2\pi$$

$$\therefore \text{Minimum value of ripple } (\delta) = \min(\delta_1, \delta_2) = 0.01$$

$$\therefore \text{Minimum Attenuation, } A = -20 \log_{10}(\delta)$$

$$= -20 \log_{10}(0.01)$$

$$= 40$$

for Kaiser window,

$$\beta = \begin{cases} 0.1102 (A - 8.7) & ; A > 50 \\ 0.5842 (A - 21)^{0.4} + 0.07886 (A - 21) & ; 21 \leq A \leq 50 \\ 0 & ; A < 21 \end{cases}$$

Since  $A = 40 (< 50)$ , so,

$$\begin{aligned} \beta &= 0.5842 (A - 21)^{0.4} + 0.07886 (A - 21) \\ &= 0.5842 (40 - 21)^{0.4} + 0.07886 (40 - 21) \end{aligned}$$

$$\therefore \beta = 3.3953$$

Also,

$$M = \frac{A - 8}{2.285 \Delta \omega} = \frac{40 - 8}{2.285 \times 0.02\pi} = 222.88$$

Since  $M$  must be in even, so

$$M \approx 224$$

To obtain Kaiser window,

$$\alpha = \frac{M}{2} = \frac{224}{2} = 112$$

for Kaiser window,

$$w(n) = \begin{cases} I_0 \left\{ \beta \left( 1 - \left( \frac{n - \alpha}{\alpha} \right)^2 \right)^{1/2} \right\} & ; 0 \leq n \leq M \\ I_0 \beta & \\ 0 & ; \text{otherwise} \end{cases}$$

$$\text{where, } I_0(x) = 1 + \frac{0.25x^2}{(1!)^2} + \frac{(0.25x^2)^2}{(2!)^2} + \dots$$

To obtain unit impulse response  $h_d(n)$ .  
For low pass filter, the desired frequency response is;

$$H_d(\omega) = \begin{cases} e^{-j\omega\alpha} & ; -\omega_c \leq \omega \leq \omega_c \\ 0 & ; \text{otherwise} \end{cases}$$

On solving by taking the inverse Fourier Transform of  $H_d(\omega)$ , we get;

$$h_d(n) = \begin{cases} \frac{\sin \omega_c(n-\alpha)}{\omega_c \pi(n-\alpha)} & ; \text{for } n \neq \alpha \\ \frac{\omega_c \pi(n-\alpha)}{\pi} & ; n = \alpha \end{cases}$$

Since  $\alpha = \frac{M}{2} = 112$  and  $\omega_c = 0.2\pi$

$$h_d(n) = \begin{cases} \frac{\sin 0.2\pi(n-112)}{\pi(n-112)} & ; n \neq 112 \\ 0.2 & ; n = \alpha \end{cases}$$

Now,

$$h(n) = h_d(n) \cdot \omega(n)$$

$$\therefore h(n) = \begin{cases} \frac{\sin 0.2\pi(n-112)}{\pi(n-112)} \times \frac{I_0 \left\{ 3.3953 \left( 1 - \left( \frac{n-112}{112} \right)^2 \right)^{1/2} \right\}}{I_0(3.3953)} & ; n \neq 112 \\ 0.2 \times \frac{I_0 \left\{ 3.3953 \left( 1 - \left( \frac{n-112}{112} \right)^2 \right)^{1/2} \right\}}{I_0(3.3953)} & ; n = 112 \end{cases}$$

$$\therefore h(n) = \begin{cases} \frac{\sin 0.2\pi(n-112)}{\pi(n-112)} \times \frac{I_0 \left\{ 3.3953 \left( 1 - \left( \frac{n-112}{112} \right)^2 \right)^{1/2} \right\}}{I_0(3.3953)} & ; n \neq 112 \\ 0.2 \times \frac{I_0 \left\{ 3.3953 \left( 1 - \left( \frac{n-112}{112} \right)^2 \right)^{1/2} \right\}}{I_0(3.3953)} & ; n = 112 \end{cases}$$

This is the required relationship of unit sample response of FIR filter using Kaiser window.

5. b) Design a digital low pass filter using Bilinear Transformation to satisfy the following requirements
- Monotonic stopband and passband
  - -3dB cut off frequency at  $0.6\pi$  radians and
  - Magnitude down at 15 dB at  $0.75\pi$  radians

sol:-

6.a) Convert the analog filter with system function,

$$H_a(s) = \frac{s + 0.2}{(s + 0.2)^2 + 9}$$

into a digital filter by means of bilinear transformation. The digital filter is to have a resonant frequency of  $\omega_r = \pi/2$ .

~~sol~~ The given system function is:

$$H_a(s) = \frac{s + 0.2}{(s + 0.2)^2 + 9} = \frac{s + 0.2}{(s + 0.2)^2 + 3^2} \quad \text{--- (1)}$$

from (1), it is clear that  $\Omega = 3$

The value of  $\omega_r = \frac{\pi}{2}$

We know,

$$\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$$

$$\Rightarrow 3 = \frac{2}{T} \tan\left(\frac{\pi/2}{2}\right)$$

$$\Rightarrow T = \frac{2}{3} \tan\left(\frac{\pi}{4}\right)$$

$$\therefore T = \frac{2}{3} \text{ sec}$$

i.e. Sampling Time,  $T = 2/3 \text{ sec}$ .

for Bilinear transformation,

$$s = \frac{2}{T} \left( \frac{z-1}{z+1} \right)$$

From (1),

$$H(z) = \frac{\frac{2}{3} \left( \frac{z-1}{z+1} \right) + 0.2}{\left( \frac{2}{3} \left( \frac{z-1}{z+1} \right) + 0.2 \right)^2 + 9}$$

$$= \frac{3 \left( \frac{z-1}{z+1} \right) + 0.2}{\left( 3 \left( \frac{z-1}{z+1} \right) + 0.2 \right)^2 + 9}$$

$$= \frac{(3z-3) + 0.2}{\left( \frac{3z-3}{z+1} + 0.2 \right)^2 + 9}$$

$$= \frac{(3z-3) + 0.2z + 0.2}{z+1}$$

$$= \frac{(3z-3+0.2z+0.2)^2 + 9}{(z+1)^2}$$

$$= \frac{3.2z - 2.8}{z+1}$$

$$= \frac{(3.2z - 2.8)^2 + 9(z+1)^2}{(z+1)^2}$$

$$= \frac{(3.2z - 2.8)(z+1)}{(3.2z - 2.8)^2 + 9z^2 + 18z + 9}$$

$$= \frac{(3.2z - 2.8)(z+1)}{9z^2 + 18z + 9}$$

$$= \frac{(3.2z - 2.8)(z+1)}{9z^2 + 18z + 9}$$

$$= \frac{(3.2z - 2.8)(z+1)}{9z^2 + 18z + 9}$$

$$= \frac{3.2z^2 + 3.2z - 2.8z - 2.8}{19.24z^2 + 0.08z + 16.84}$$

$$\therefore H(z) = \frac{3.2z^2 + 0.4z - 2.8}{19.24z^2 + 0.08z + 16.84}$$

This is the required transfer function for digital IIR filter Ans:

6.b) Differentiate between FIR and IIR digital filters with examples.

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|       | IIR Filter                                                              | FIR Filter                                                           |
|-------|-------------------------------------------------------------------------|----------------------------------------------------------------------|
| (i)   | IIR filter stands for Infinite Impulse Response filter.                 | FIR stands for Finite Impulse Response Filter.                       |
| (ii)  | IIR filter provides impulse responses for an infinite duration of time. | FIR filter provides impulse responses for a finite duration of time. |
| (iii) | In <b>IIR</b> filters, feedback system is available.                    | In <b>FIR</b> filters, feedback system is not available.             |
| (iv)  | IIR filter is recursive in nature.                                      | FIR filter is non-recursive in nature.                               |
| (v)   | IIR filters are less stable due to the presence of poles.               | FIR filters are always stable due to absence of poles.               |
| (vi)  | The transfer function of IIR filter contains both poles and zeros.      | The transfer function of FIR filter contains only zeros.             |

|                                                                                                                          |                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| (vii) Due to presence of feedback, IIR filter requires less memory.                                                      | Due to absence of feedback system, FIR filter requires more memory.                                       |
| (viii) The computational efficiency of IIR filter is more.                                                               | The computational efficiency of FIR filter is less.                                                       |
| (ix) IIR filters are used in biomedical sensor, telecommunication, audio equalization etc.                               | FIR filters are used in audio system, control system, medical devices etc.                                |
| (x) For example:<br>A 3-point moving average FIR filter.<br>$y(n) = \frac{1}{3} \{ x(n) + x(n-1) + x(n-2) + \dots \}$    | For example:<br>A simple first-order low-pass filter I<br>$y(n) = \dots$                                  |
| (xi) For example:<br>A simple first-order low-pass IIR filter,<br>$y(n) = a \cdot y(n-1) + (1-a) \cdot x(n) ; 0 < a < 1$ | For example:<br>A 3-point moving average FIR filter,<br>$y(n) = \frac{1}{3} \{ x(n) + x(n-1) + x(n-2) \}$ |

7. Write short notes on:

(a) Linear Convolution vs. Circular Convolution;

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Linear Convolution

Circular Convolution

(i) Linear convolution is the multiplication of two sequences in time domain.

Circular convolution is the multiplication of two sequences ~~seq~~ in frequency domain.

(ii) The output length of ~~cir~~ linear convolution is  $L+M-1$ .

The output length of circular convolution is  $N$  (i.e. same as DFT size)

(iii) Zero padding is not required in linear convolution.

Zero padding is required in circular convolution for accurate result.

(iv) Linear convolution is used in LTI systems for real-time world applications.

Circular convolution is used in FFT-based filtering.

(v) The linear convolution is given by

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) \cdot h(n-k)$$

The circular convolution is given by

$$y(n) = \sum_{k=0}^{N-1} x(k) \cdot h((n-k)/N)$$

## b) Causal System:

Def: A discrete time system is said to be causal if the output of the system at any time  $t$  depends only on the present inputs and past inputs, not on future inputs.

$$\text{i.e. } y(n) = f[x(n), x(n-1), x(n-2), \dots]$$

If the system's output at any time  $t$  depends on the future inputs, the system is called ~~non~~ anti-causal system.

$$\therefore y(n) = f(x(n+1), x(n+2), \dots)$$

If the output of the system at any time  $t$  depends on the ~~for~~ both past and future inputs, the system is called non-causal system.

$$\therefore y(n) = f(x(n-1), x(n), x(n+1), \dots)$$

In summary,

| System Type | Depends on       | Real-Time? |
|-------------|------------------|------------|
| Causal      | Present and Past | Yes        |
| Anti-Causal | Future only      | No         |
| Non-causal  | Past & Future    | No         |